

Information constraints in fine-tuning and reasoning language models

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How much information must survive?

Fine-tuning

A compressed update

What must a fixed model receive to retain a learned behavior?

Reasoning

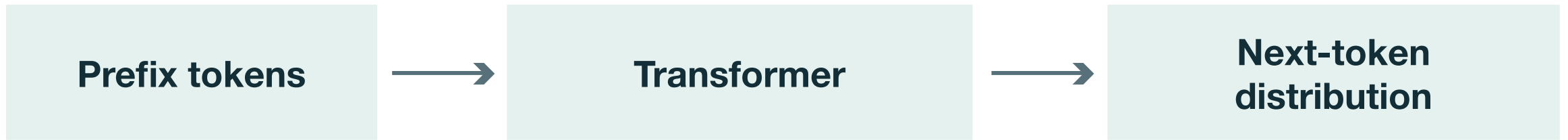
An intermediate state

What must the next computation receive to continue correctly?

Receiver, task, and update rule define what is useful.

Autoregressive language models

$$p_{\theta}(x_{1:T}) = \prod_{t=1}^T p_{\theta}(x_t \mid x_{<t})$$



Hidden vectors h_t at each layer

Training minimizes next-token negative log-likelihood.

At generation time, each new token extends the prefix.

LoRA and QLoRA

$$W' = W + \frac{\alpha}{r}BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}$$

Frozen W
4-bit backbone in QLoRA

Trainable A and B
Higher-precision adapter

$$P = r(d_{\text{in}} + d_{\text{out}}) \ll d_{\text{in}}d_{\text{out}}$$

This work compresses the adapter after training.

Count shown for one adapted matrix. Sum over modules for the full adapter.

The adapter's numerical code determines its rate

$$\widehat{\Delta W} = \frac{\alpha}{r} \widehat{B} \widehat{A}, \quad \ell_{\text{fixed}} = bP + \ell_{\text{scales}} + \ell_{\text{metadata}}$$

Before entropy coding

b bits per value, plus the information needed to decode the numbers

After packing and zlib

The actual file length determines R.
The original value count P stays fixed.

$$R = \frac{8 |\text{file}(\widehat{A}, \widehat{B})|_{\text{bytes}}}{P} \quad \text{bits per original adapter value}$$

Backbone weights remain shared. This rate counts only the complete adapter file.

Coding and rate–distortion

$$\ell(x) \simeq -\log_2 p(x), \quad H(X) = \mathbb{E}[-\log_2 p(X)]$$

Lossy coding

**Minimize the message length
under an error constraint.**

Side information

Both sides already have the base model.

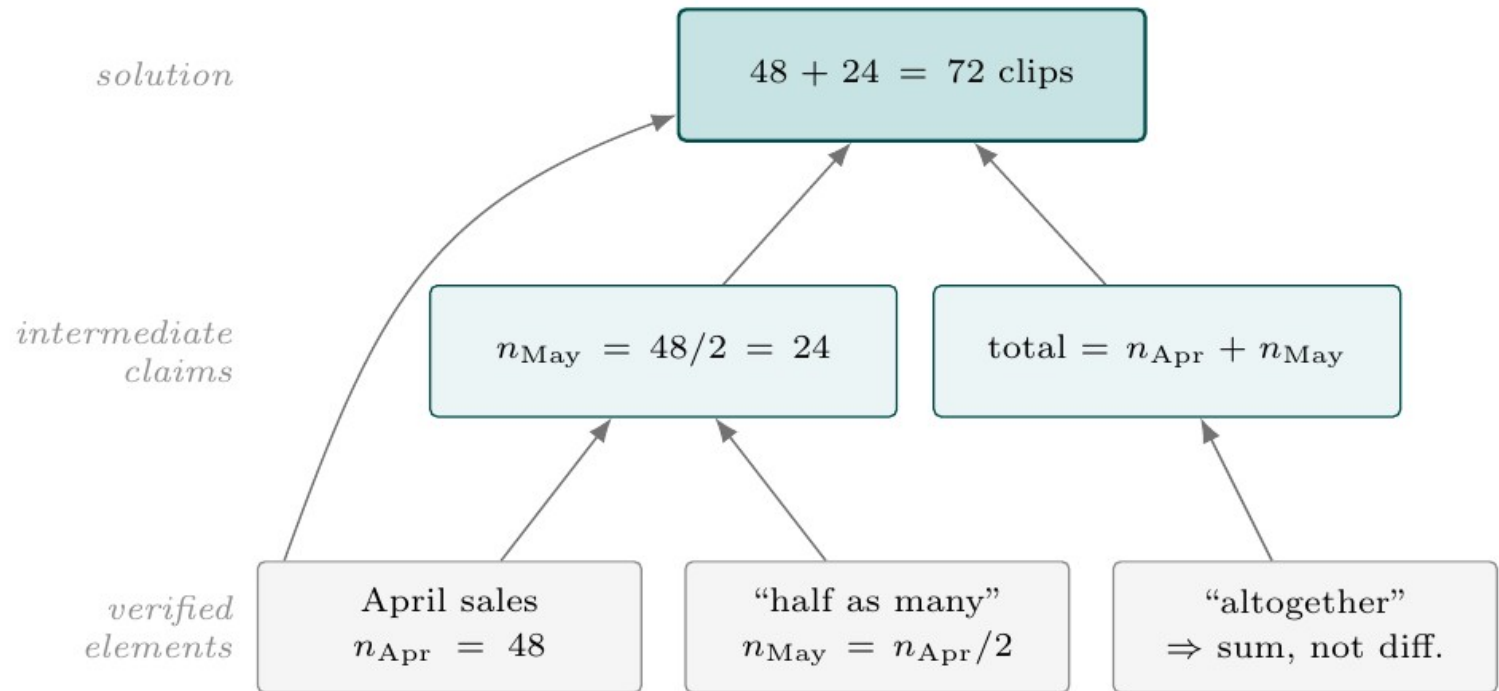
Only the update enters the file budget.

The distortion measure specifies the behavior to preserve.

A chain of thought for April and May sales

April: 48 clips
May: half as many

Total over both
months?



Intermediate claims make the computation explicit.

The report's hypothetical solution graph. Its nodes need not coincide with token or hidden-state boundaries.

How the model builds a chain of thought

$$p_{\theta}(z_{1:K}, y \mid x) = \left[\prod_{k=1}^K p_{\theta}(z_k \mid x, z_{<k}) \right] p_{\theta}(y \mid x, z_{1:K})$$

x: problem z: intermediate trace y: final answer

Generate a token, then extend the context

$$z_k \sim p_{\theta}(\cdot \mid x, z_{<k}), \quad (x, z_{<k}) \longmapsto (x, z_{\leq k})$$

Each token adds a transformer pass.
One semantic step can span many tokens.

$$p_{\theta}^{\text{CoT}}(y \mid x) = \sum_z p_{\theta}(z, y \mid x)$$

Generation samples a path through this sum.

An exact rate for a fixed learned update



$$\text{Retain}(R) = \frac{L_0 - L_R}{L_0 - L_{\text{raw}}}$$

$$R_{M,U}^*(\tau) = \min_{c \in \mathcal{C}} \{R(c) : \text{Retain}_{M,U}(c) \geq \tau\}, \quad \tau = 0.90$$

C: tested codecs M: frozen model U: scored behavior L: held-out loss

Weight error misses behavioral damage



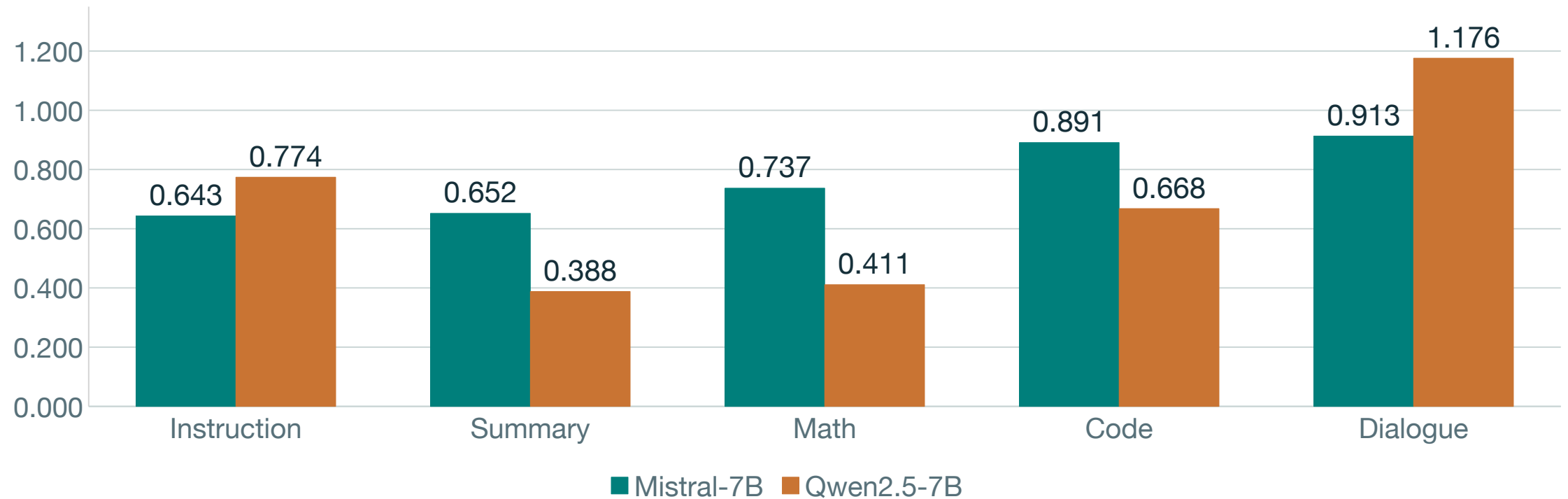
≈ 1 bit / value

Similar weight RMSE
83 points apart in
retained task gain

Behavior must be measured at the model's output.

Mistral-7B + rank-16 LoRA, MetaMathQA. Retained GSM8K exact-match gain, 3 seeds.

The corpus ordering changes with the model



$R^*(0.90)$, bits per original adapter value. Mean over 3 seeds per corpus and model.

Correction spectra are a promising predictor

$$g_t = h_t \otimes (p_t - e_{y_t}), \quad F = \frac{1}{n} G^\top G$$

0.764

Within-model Spearman ρ

$$I_{\text{corr}} = \log \det(I + F) = \sum_j \log(1 + \lambda_j(F))$$

22 conditions

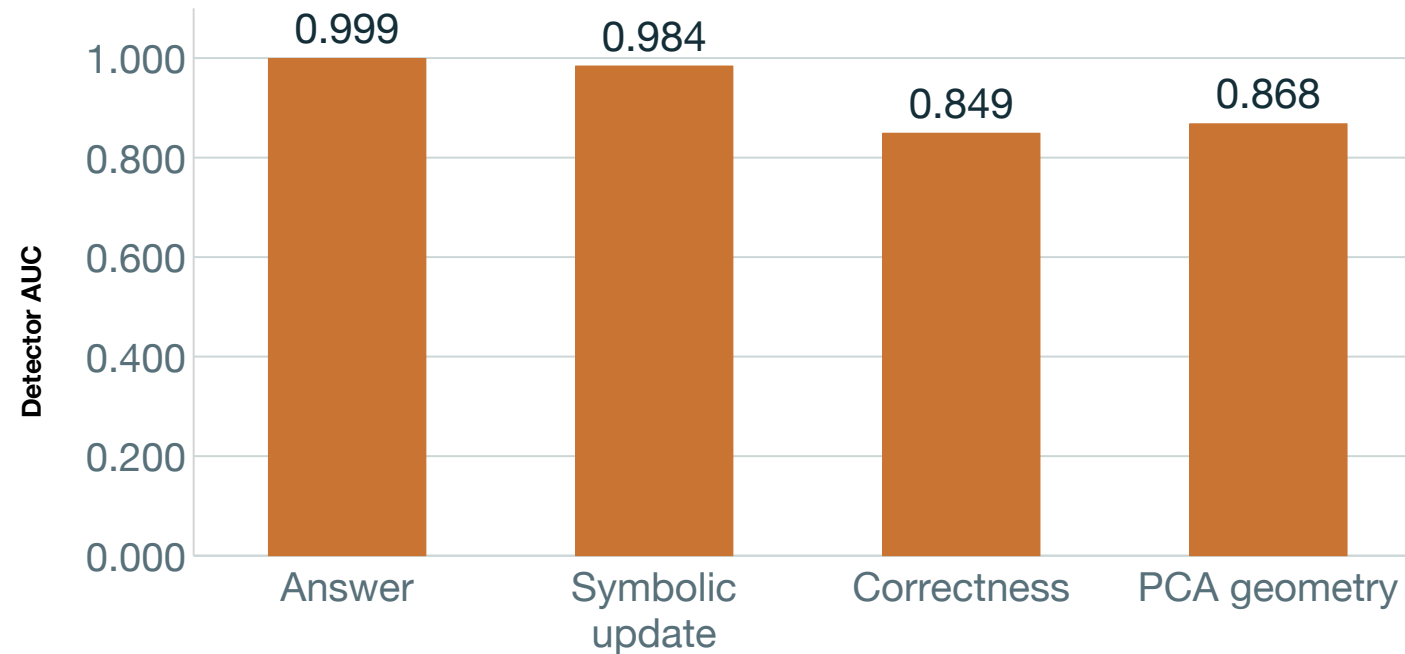
7 task families

2 frozen 7B models

Large corrections across many directions

Projected output-head gradients. Strongest within-model ordering in this screen; prospective validation remains open.

Different objectives select different boundaries



0.098–0.210

Pairwise oracle-boundary F1
within ± 4 tokens

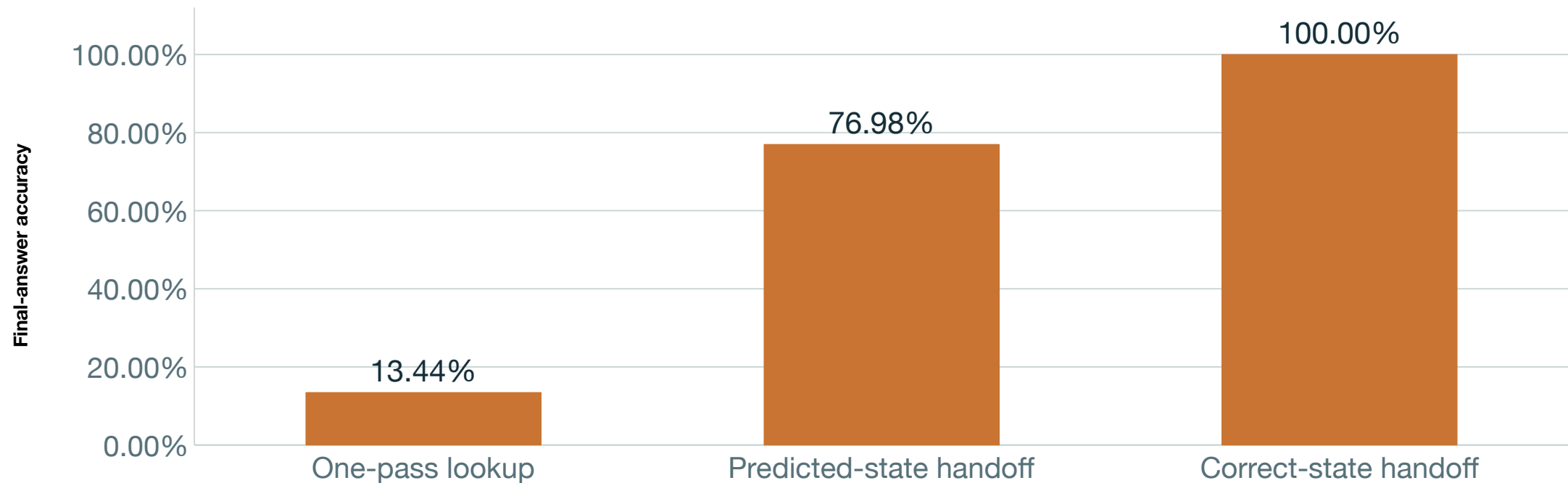
0.020

Best worst-objective utility
among tested rules

SmolLM3-3B. Equal boundary budgets. Random utility = 0, target oracle = 1.

An explicit handoff restores composition

Eight-state program, followed by a state-to-answer lookup



The fresh call receives the state after the history is deleted.

Frozen Qwen2.5-32B, horizon 2, 8 states. Chance = 12.5%. Endpoint inference = 76.98%.

Capacity and stable names matter separately

Code	Bits	Naming	Exact state
Lossy	3	Collisions required	Cannot recover all 16
Canonical	4	One name per state	100% through H = 256
Padded	5	Same stable naming	100% through H = 256
Path-aliased	5	Name depends on path	30–40% at depth 3

At five bits, naming changes state recovery.

16 proof states, matched training data and compute, 3 seeds. H is total horizon; depth counts active deductions.

Local errors accumulate under repeated use

$\approx 89\%$

Correct local transitions

7.92%

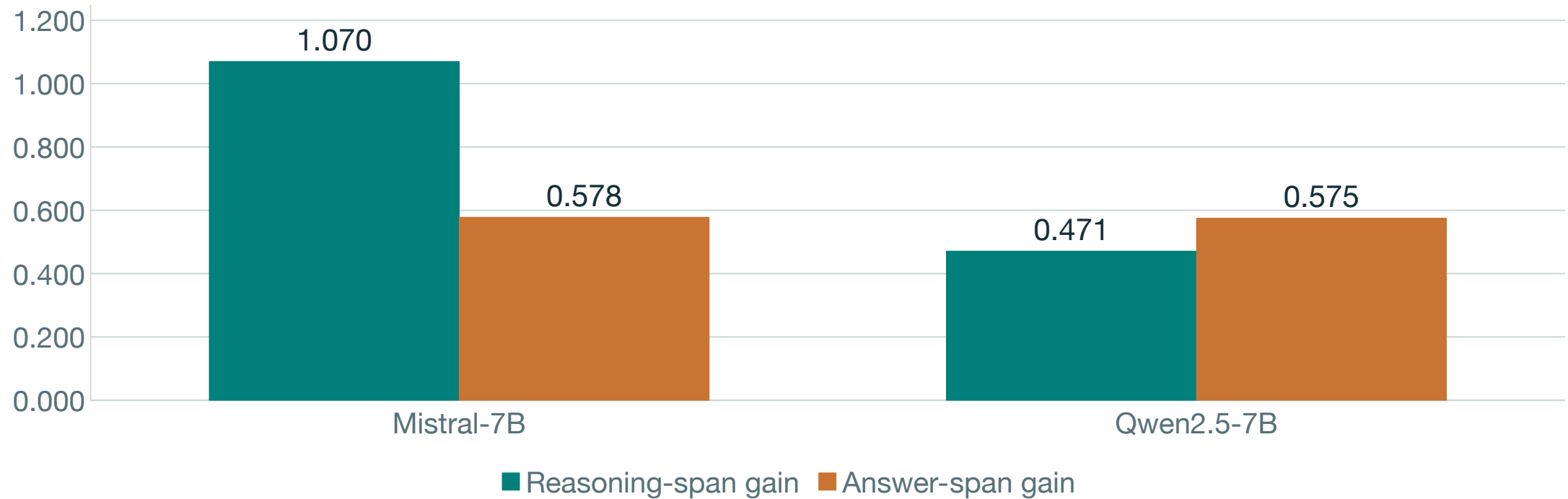
Correct final answers at $H = 32$

$$\hat{s}_{k+1} = f_{\theta}(\hat{s}_k, a_k), \quad \hat{y} = g_{\theta}(\hat{s}_K)$$

Chance: 6.25% One-pass reasoning: 5.83%

Reliable single steps do not ensure reliable long executions.

Reasoning and answer rates also reverse



$R^*(0.90)$, bits per adapter value. MetaMathQA. This measures storage for retained next-token gain.

What the experiments establish

Fine-tuning

**A measured file rate
for a stated retained behavior**

The receiver changes the cost.

Reasoning

**A state must support
its intended next
computation**

Repeated use adds an update
requirement.

A common interpretation; a predictive law remains open.

The next tests must make new predictions

01 Predict before training

Use span-specific correction spectra on new models and tasks.
Freeze predictions of R^* before observing the adapter rates.

02 Follow state use along the rate ladder

Measure decoding, handoff, and answer behavior
while compressing the same reasoning adapter.

Useful information depends on its use

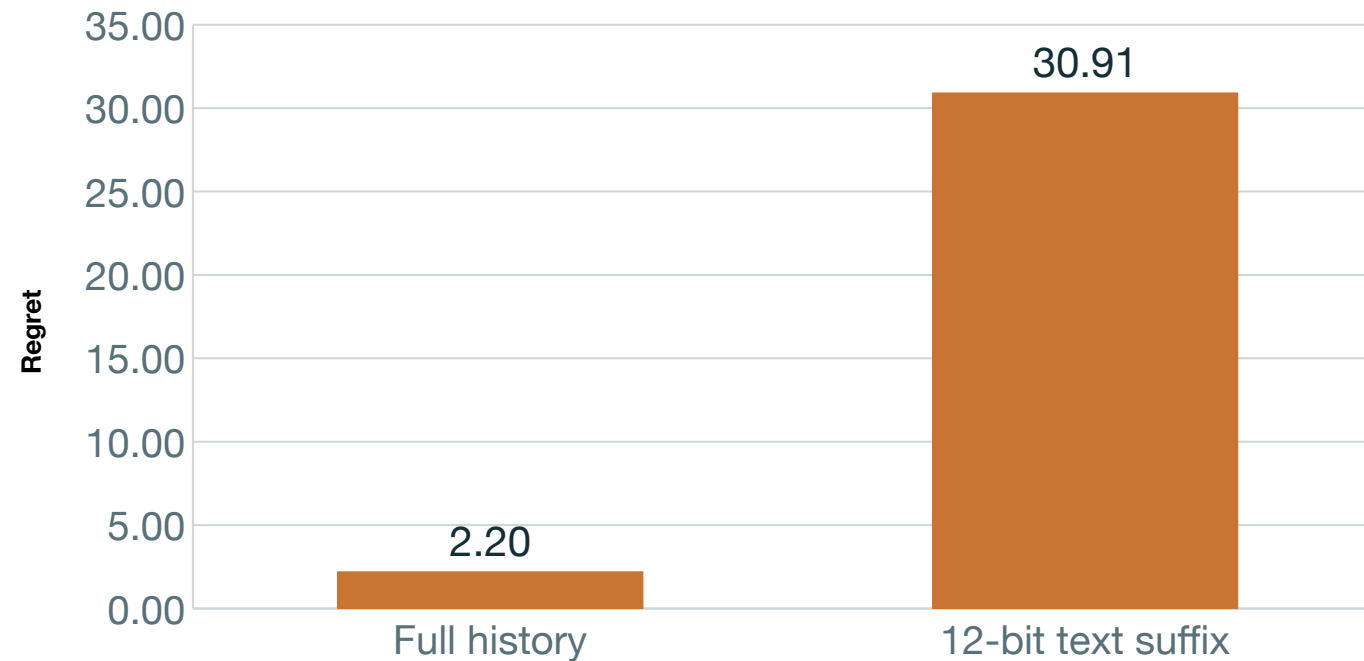
Receiver

Which model must use the information?

Task and update rule

Which behavior must survive, and for how long?

Equal snapshot information, unequal memory

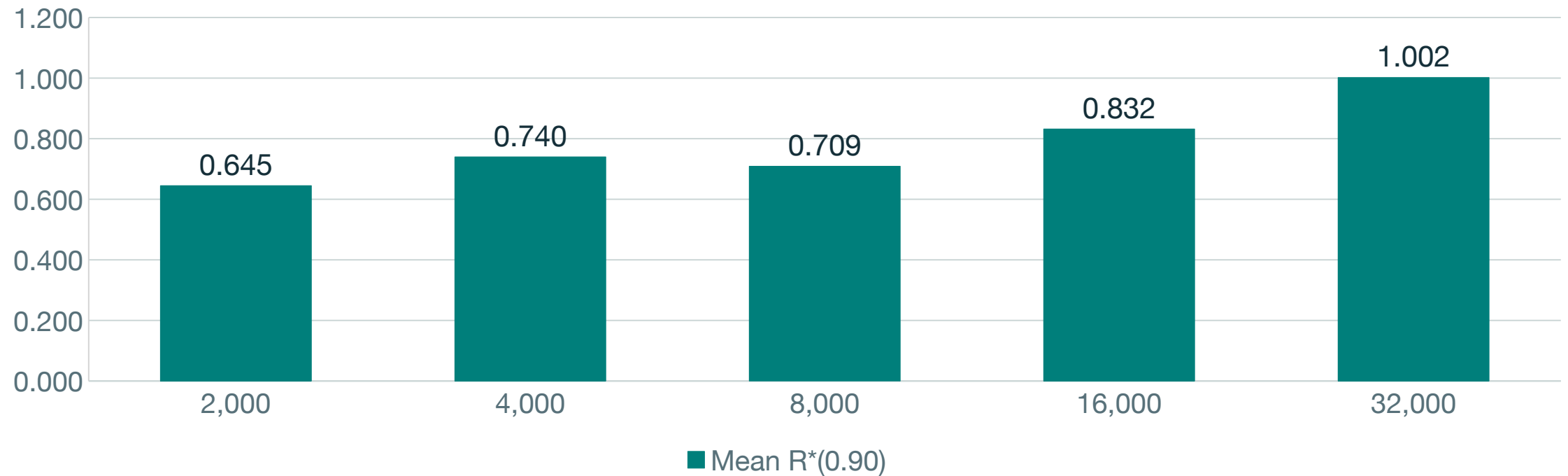


Same initial information

$I(\text{state}; \text{memory}) = 2.512$ bits
Path leakage = 1.781 bits

Exact finite control, 90 task cells. Regret is lower-is-better. This is separate from the LLM assays.

Distinct examples at fixed training exposure



Distinct training examples. 2,000 optimizer steps and 32,000 presentations fixed. 3 seeds per condition.

What the codec and threshold count

Included in the file

Quantized factors
Scales and metadata
Bit-packed, zlib-compressed payload

Conditioned on

A shared frozen backbone
One fixed adapter checkpoint
A named held-out utility

R^* is an attainable rate within the tested codec family.

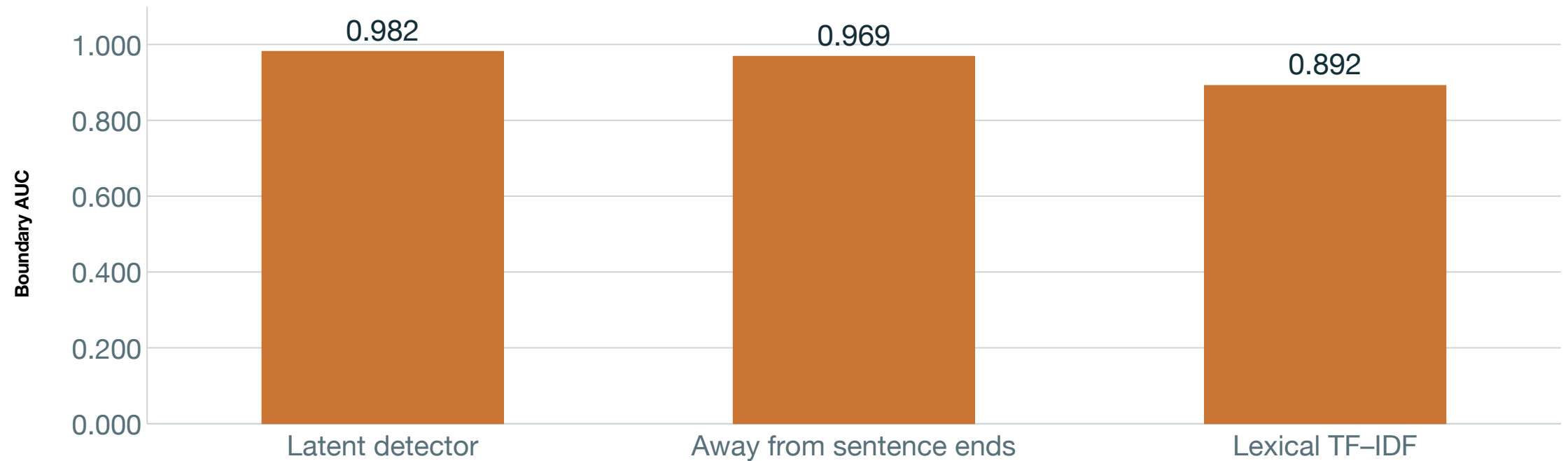
Sub-bit rates use paired rank-direction removal. File rate uses the original adapter value count.

Prediction, accuracy, and state recovery

Metric	Question
Held-out NLL	Does the model predict the scored tokens?
Retained gain	How much improvement over the base survives?
Answer exact match	Is the final task answer correct?
Exact state	Does the complete supplied ledger match?
Boundary AUC / F1	Can we detect a target / do selected gaps agree?

Every reported rate and retention value needs a named utility and evaluation span.

Semantic boundaries are decodable too



1,761 judge-labeled spans, 58 traces. Question-disjoint evaluation. Labels come from Qwen-110B-A25B.

Correction-spectrum screen and its limits

Measure	Within-model ρ	Adjusted p	Held-out RMSE
Correction log-volume	0.764	0.00034	0.187
Correction effective rank	0.609	0.0112	0.181
Surprise-weighted log-volume	0.614	0.0103	0.204
Base target code length	0.200	0.524	0.261
Model-only fit	—	—	0.264

RMSE in bits per adapter value, lower is better. Family-held-out evaluation. 22 conditions, 10 screened measures.